Question ID:101461
Topic Name: Mathematics-Section A
Question:

Let a circle C touch the lines $L_1 : 4x-3y+K_1=0$ and $L_2 : 4x-3y+K_2=0$, $K_1, K_2 \in \mathbf{R}$. If a line passing through the centre of the circle C intersects L_1 at $(-1, 2)$ and L_2 at $(3, -6)$, then the equation of the circle C is :

A $(x-1)^2+(y-2)^2=4$

B $(x+1)^2+(y-2)^2=4$

C $(x-1)^2+(y+2)^2=16$

D $(x-1)^2+(y-2)^2=16$

Answer Given By Candidate:**A**

Question ID:101462
Topic Name: Mathematics-Section A
Question:

The value of $\int_0^{\pi} \frac{e^{\cos x}\sin x}{(1+\cos^2 x)(e^{\cos x}+e^{-\cos x})}\,dx$ is equal to :

A $\frac{\pi^2}{4}$

B $\frac{\pi^2}{2}$

C $\frac{\pi}{4}$

D $\frac{\pi}{2}$

Answer Given By Candidate:**A**

Question ID:101463
Topic Name: Mathematics-Section A

Question:

Let a, b and c be the length of sides of a triangle ABC such that $\frac{a+b}{7} = \frac{b+c}{8} = \frac{c+a}{9}$

If r and R are the radius of incircle and radius of circumcircle of the triangle ABC, respectively, then the value of $\frac{R}{r}$ is equal to :

A $\frac{5}{2}$

B 2

C $\frac{3}{2}$

D 1

Answer Given By Candidate:**Not Attempted**

Question ID:101464
Topic Name: Mathematics-Section A
Question:

Let $f : \mathbf{N} \to \mathbf{R}$ be a function such that $f(x+y) = 2f(x)\, f(y)$ for natural numbers x and y. If $f(1) = 2$, then the value of α for which

$$\sum_{k=1}^{10} f(\alpha + k) = \frac{512}{3}(2^{20} - 1)$$

holds, is :

A 2

B 3

C 4

D 6

Answer Given By Candidate:**B**

Question ID:101465
Topic Name: Mathematics-Section A

Question:

Let A be a 3×3 real matrix such that

$$A\begin{pmatrix}1\\1\\0\end{pmatrix}=\begin{pmatrix}1\\1\\0\end{pmatrix};A\begin{pmatrix}1\\0\\1\end{pmatrix}=\begin{pmatrix}-1\\0\\1\end{pmatrix}\text{ and }A\begin{pmatrix}0\\0\\1\end{pmatrix}=\begin{pmatrix}1\\1\\2\end{pmatrix}.$$

If $X=(x_1, x_2, x_3)^T$ and I is an identity matrix of order 3, then the system $(A-2I)X=\begin{pmatrix}4\\1\\1\end{pmatrix}$ has :

A no solution

B infinitely many solutions

C unique solution

D exactly two solutions

Answer Given By Candidate:**Not Attempted**

Question ID:101466
Topic Name: Mathematics-Section A
Question:

Let $f:\mathbf{R}\to\mathbf{R}$ be defined as

$$f(x)=x^3+x-5$$

If $g(x)$ is a function such that $f(g(x))=x,\ \forall\ 'x'\in\mathbf{R}$, then $g'(63)$ is equal to _____.

A $\frac{1}{49}$

B $\frac{3}{49}$

C $\frac{43}{49}$

D $\frac{91}{49}$

Answer Given By Candidate:**A**

Question ID:101467
Topic Name: Mathematics-Section A

Consider the following two propositions :

P1 : $\sim(p \rightarrow \sim q)$

P2 : $(p \wedge \sim q) \wedge ((\sim p) \vee q)$

If the proposition $p \rightarrow ((\sim p) \vee q)$ is evaluated as FALSE, then :

Question:

A P1 is TRUE and P2 is FALSE

B P1 is FALSE and P2 is TRUE

C Both P1 and P2 are FALSE

D Both P1 and P2 are TRUE

Answer Given By Candidate:**Not Attempted**

Question ID:101468
Topic Name: Mathematics-Section A
Question:

If $\frac{1}{2 \cdot 3^{10}} + \frac{1}{2^2 \cdot 3^9} + \ldots + \frac{1}{2^{10} \cdot 3} = \frac{K}{2^{10} \cdot 3^{10}}$, then the remainder when K is divided by 6 is :

A 1

B 2

C 3

D 5

Answer Given By Candidate:**A**

Question ID:101469
Topic Name: Mathematics-Section A
Question:

Let $f(x)$ be a polynomial function such that $f(x)+f'(x)+f''(x)=x^5+64$. Then, the value of $\lim\limits_{x \to 1} \frac{f(x)}{x-1}$ is equal to :

A -15

B -60

C 60

D 15

Answer Given By Candidate:**Not Attempted**

Question ID:101470
Topic Name: Mathematics-Section A
Question:

Let E_1 and E_2 be two events such that the conditional probabilities $P(E_1|E_2) = \frac{1}{2}$, $P(E_2|E_1) = \frac{3}{4}$ and $P(E_1 \cap E_2) = \frac{1}{8}$. Then :

A $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$

B $P(E_1' \cap E_2') = P(E_1') \cdot P(E_2)$

C $P(E_1 \cap E_2') = P(E_1) \cdot P(E_2)$

D $P(E_1' \cap E_2) = P(E_1) \cdot P(E_2)$

Answer Given By Candidate:C

Question ID:101471
Topic Name: Mathematics-Section A
Question:

Let $A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$. If M and N are two matrices given by $M = \sum_{k=1}^{10} A^{2k}$ and $N = \sum_{k=1}^{10} A^{2k-1}$ then MN^2 is :

A a non-identity symmetric matrix

B a skew-symmetric matrix

C neither symmetric nor skew-symmetric matrix

D an identity matrix

Answer Given By Candidate:Not Attempted

Question ID:101472
Topic Name: Mathematics-Section A
Question:

Let $g : (0, \infty) \to \mathbf{R}$ be a differentiable function such that

$$\int \left(\frac{x(\cos x - \sin x)}{e^x + 1} + \frac{g(x)\left(e^x + 1 - xe^x\right)}{(e^x + 1)^2} \right) dx = \frac{x\, g(x)}{e^x + 1} + c, \text{ for all } x > 0, \text{ where c is an}$$

arbitrary constant. Then :

A g is decreasing in $\left(0, \frac{\pi}{4}\right)$

B g' is increasing in $\left(0, \frac{\pi}{4}\right)$

C $g + g'$ is increasing in $\left(0, \frac{\pi}{2}\right)$

D $g - g'$ is increasing in $\left(0, \frac{\pi}{2}\right)$

Answer Given By Candidate:Not Attempted

Question ID:101473
Topic Name: Mathematics-Section A
Question:

Let $f : \mathbf{R} \to \mathbf{R}$ and $g : \mathbf{R} \to \mathbf{R}$ be two functions defined by $f(x) = \log_e(x^2+1) - e^{-x} + 1$ and $g(x) = \dfrac{1 - 2e^{2x}}{e^x}$. Then, for which of the following range of α, the inequality

$$f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right) \text{ holds ?}$$

A $(2, 3)$

B $(-2, -1)$

C $(1, 2)$

D $(-1, 1)$

Answer Given By Candidate:**Not Attempted**

Question ID:101474
Topic Name: Mathematics-Section A
Question:

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ $a_i > 0$, $i = 1, 2, 3$ be a vector which makes equal angles with the coordinate axes OX, OY and OZ. Also, let the projection of $\vec{a}$ on the vector $3\hat{i} + 4\hat{j}$ be 7.

Let $\vec{b}$ be a vector obtained by rotating $\vec{a}$ with 90°. If $\vec{a}$, $\vec{b}$ and x-axis are coplanar, then projection of a vector $\vec{b}$ on $3\hat{i} + 4\hat{j}$ is equal to :

A $\sqrt{7}$

B $\sqrt{2}$

C 2

D 7

Answer Given By Candidate:**Not Attempted**

Question ID:101475
Topic Name: Mathematics-Section A
Question:

Let $y = y(x)$ be the solution of the differential equation $(x+1)y' - y = e^{3x}(x+1)^2$, with $y(0) = \dfrac{1}{3}$.

Then, the point $x = -\dfrac{4}{3}$ for the curve $y = y(x)$ is :

A not a critical point

B a point of local minima

C a point of local maxima

D a point of inflection

Answer Given By Candidate:B

Question ID:101476
Topic Name: Mathematics-Section A
Question:

If $y=m_1x+c_1$ and $y=m_2x+c_2$, $m_1 \neq m_2$ are two common tangents of circle $x^2+y^2=2$ and parabola $y^2=x$, then the value of $8|m_1m_2|$ is equal to :

A $3+4\sqrt{2}$

B $-5+6\sqrt{2}$

C $-4+3\sqrt{2}$

D $7+6\sqrt{2}$

Answer Given By Candidate:Not Attempted

Question ID:101477
Topic Name: Mathematics-Section A
Question:

Let Q be the mirror image of the point P(1, 0, 1) with respect to the plane S : $x+y+z=5$. If a line L passing through $(1, -1, -1)$, parallel to the line PQ meets the plane S at R, then QR^2 is equal to :

A 2

B 5

C 7

D 11

Answer Given By Candidate:Not Attempted

Question ID:101478
Topic Name: Mathematics-Section A
Question:

If the solution curve $y=y(x)$ of the differential equation $y^2dx+(x^2-xy+y^2)dy=0$, which passes through the point $(1, 1)$ and intersects the line $y=\sqrt{3}x$ at the point $(\alpha, \sqrt{3}\alpha)$, then value of $\log_e(\sqrt{3}\alpha)$ is equal to :

A $\frac{\pi}{3}$

B $\frac{\pi}{2}$

C

$\frac{\pi}{12}$

D

$\frac{\pi}{6}$

Answer Given By Candidate:**B**

Question ID:101479
Topic Name: Mathematics-Section A
Question:

Let $x=2t$, $y=\frac{t^2}{3}$ be a conic. Let S be the focus and B be the point on the axis of the conic such that SA $\perp$ BA, where A is any point on the conic. If k is the ordinate of the centroid of the ΔSAB, then $\lim_{t\to 1} k$ is equal to :

A

$\frac{17}{18}$

B

$\frac{19}{18}$

C

$\frac{11}{18}$

D

$\frac{13}{18}$

Answer Given By Candidate:**Not Attempted**

Question ID:101480
Topic Name: Mathematics-Section A
Question:

Let a circle C in complex plane pass through the points $z_1=3+4i$, $z_2=4+3i$ and $z_3=5i$. If $z(\neq z_1)$ is a point on C such that the line through z and z_1 is perpendicular to the line through z_2 and z_3, then arg(z) is equal to :

A

$\tan^{-1}\left(\frac{2}{\sqrt{5}}\right)-\pi$

B

$\tan^{-1}\left(\frac{24}{7}\right)-\pi$

C

$\tan^{-1}(3)-\pi$

D

$\tan^{-1}\left(\frac{3}{4}\right)-\pi$

Answer Given By Candidate:**Not Attempted**

Question ID:101481
Topic Name: Mathematics-Section B
Question:

Let C_r denote the binomial coefficient of x^r in the expansion of $(1+x)^{10}$. If for $\alpha, \beta \in \mathbf{R}$,

$$C_1 + 3 \cdot 2\, C_2 + 5 \cdot 3\, C_3 + \ldots \text{upto 10 terms} = \frac{\alpha \times 2^{11}}{2^{\beta} - 1}\left(C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots \text{upto 10 terms}\right)$$

then the value of $\alpha + \beta$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101482
Topic Name: Mathematics-Section B
Question:

The number of 3-digit odd numbers, whose sum of digits is a multiple of 7, is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101483
Topic Name: Mathematics-Section B
Question:

Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$, where $|\vec{a}| = 4$, $|\vec{b}| = 3$ and $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$.

Then

$$\left|(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b})\right|^2 + 4(\vec{a} \cdot \vec{b})^2 \text{ is equal to } _____.$$

Answer Given By Candidate:**Not Attempted**

Question ID:101484
Topic Name: Mathematics-Section B
Question:

Let the abscissae of the two points P and Q be the roots of $2x^2 - rx + p = 0$ and the ordinates of P and Q be the roots of $x^2 - sx - q = 0$. If the equation of the circle described on PQ as diameter is $2(x^2 + y^2) - 11x - 14y - 22 = 0$, then $2r + s - 2q + p$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101485
Topic Name: Mathematics-Section B
Question:

The number of values of x in the interval $\left(\frac{\pi}{4}, \frac{7\pi}{4}\right)$ for which $14\text{cosec}^2 x - 2\sin^2 x = 21 - 4\cos^2 x$ holds, is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101486
Topic Name: Mathematics-Section B
Question:

For a natural number n, let $\alpha_n = 19^n - 12^n$. Then, the value of $\frac{31\alpha_9 - \alpha_{10}}{57\alpha_8}$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101487
Topic Name: Mathematics-Section B
Question:

Let $f: \mathbf{R} \to \mathbf{R}$ be a function defined by $f(x) = \left(2\left(1 - \frac{x^{25}}{2}\right)\left(2 + x^{25}\right)\right)^{\frac{1}{50}}$. If the function $g(x) = f\left(f\left(f(x)\right)\right) + f\left(f(x)\right)$, then the greatest integer less than or equal to $g(1)$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101488
Topic Name: Mathematics-Section B
Question:

Let the lines

$$L_1 : \vec{r} = \lambda\left(\hat{i} + 2\hat{j} + 3\hat{k}\right), \lambda \in \mathbf{R}$$

$$L_2 : \vec{r} = \left(\hat{i} + 3\hat{j} + \hat{k}\right) + \mu\left(\hat{i} + \hat{j} + 5\hat{k}\right); \mu \in \mathbf{R},$$

intersect at the point S. If a plane $ax + by - z + d = 0$ passes through S and is parallel to both the lines L_1 and L_2, then the value of $a + b + d$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101489
Topic Name: Mathematics-Section B
Question:

Let A be a 3×3 matrix having entries from the set $\{-1, 0, 1\}$. The number of all such matrices A having sum of all the entries equal to 5, is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101490
Topic Name: Mathematics-Section B
Question:

The greatest integer less than or equal to the sum of first 100 terms of the sequence $\frac{1}{3}, \frac{5}{9}, \frac{19}{27}, \frac{65}{81}, \cdots$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101401
Topic Name: Physics-Section A
Question:

If $Z = \frac{A^2B^3}{C^4}$, then the relative error in Z will be :

A

$\frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta C}{C}$

B $\frac{2\,\Delta A}{A}+\frac{3\,\Delta B}{B}-\frac{4\,\Delta C}{C}$

C $\frac{2\,\Delta A}{A}+\frac{3\,\Delta B}{B}+\frac{4\,\Delta C}{C}$

D $\frac{\Delta A}{A}+\frac{\Delta B}{B}-\frac{\Delta C}{C}$

Answer Given By Candidate:C

Question ID:101402
Topic Name: Physics-Section A
Question:

$\vec{A}$ is a vector quantity such that $|\vec{A}|$ = non - zero constant. Which of the following expression is true for $\vec{A}$?

A $\vec{A}\cdot\vec{A}=0$

B $\vec{A}\times\vec{A}<0$

C $\vec{A}\times\vec{A}=0$

D $\vec{A}\times\vec{A}>0$

Answer Given By Candidate:C

Question ID:101403
Topic Name: Physics-Section A
Question:

Which of the following relations is true for two unit vector $\hat{A}$ and $\hat{B}$ making an angle θ to each other ?

A $|\hat{A}+\hat{B}|=|\hat{A}-\hat{B}|\tan\theta/2$

B $|\hat{A}-\hat{B}|=|\hat{A}+\hat{B}|\tan\theta/2$

C $|\hat{A}+\hat{B}|=|\hat{A}-\hat{B}|\cos\theta/2$

D $|\hat{A}-\hat{B}|=|\hat{A}+\hat{B}|\cos\theta/2$

Answer Given By Candidate:Not Attempted

Question ID:101404
Topic Name: Physics-Section A

Question:

If force $\vec{F} = 3\hat{i} + 4\hat{j} - 2\hat{k}$ acts on a particle having position vector $2\hat{i} + \hat{j} + 2\hat{k}$ then, the torque about the origin will be :

A $3\hat{i} + 4\hat{j} - 2\hat{k}$

B $-10\hat{i} + 10\hat{j} + 5\hat{k}$

C $10\hat{i} + 5\hat{j} - 10\hat{k}$

D $10\hat{i} + \hat{j} - 5\hat{k}$

Answer Given By Candidate:**B**

Question ID:101405
Topic Name: Physics-Section A
Question:

The height of any point P above the surface of earth is equal to diameter of earth. The value of acceleration due to gravity at point P will be : (Given g = acceleration due to gravity at the surface of earth).

A $g/2$

B $g/4$

C $g/3$

D $g/9$

Answer Given By Candidate:**B**

Question ID:101406
Topic Name: Physics-Section A
Question:

The terminal velocity (v_t) of the spherical rain drop depends on the radius (r) of the spherical rain drop as :

A $r^{1/2}$

B r

C r^2

D r^3

Answer Given By Candidate:**A**

Question ID:101407
Topic Name: Physics-Section A
Question:

The relation between root mean square speed (v_{rms}) and most probable speed (v_p) for the molar mass M of oxygen gas molecule at the temperature of 300 K will be :

A

$v_{rms} = \sqrt{\frac{2}{3}}\, v_p$

B $v_{rms} = \sqrt{3/2}\, v_p$

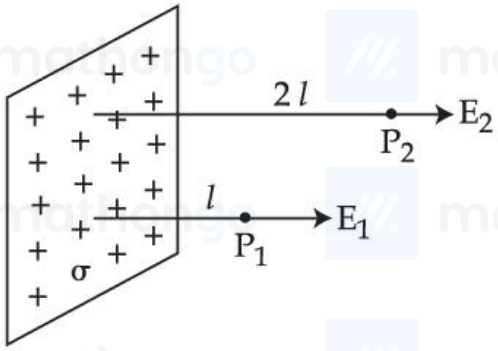
C $v_{rms} = v_p$

D $v_{rms} = \sqrt{1/3}\, v_p$

Answer Given By Candidate:**B**

Question ID:101408
Topic Name: Physics-Section A
Question:

In the figure, a very large plane sheet of positive charge is shown. P_1 and P_2 are two points at distance l and $2l$ from the charge distribution. If σ is the surface charge density, then the magnitude of electric fields E_1 and E_2 at P_1 and P_2 respectively are :



A $E_1 = \sigma/\varepsilon_o$, $E_2 = \sigma/2\varepsilon_o$

B $E_1 = 2\sigma/\varepsilon_o$, $E_2 = \sigma/\varepsilon_o$

C $E_1 = E_2 = \sigma/2\varepsilon_o$

D $E_1 = E_2 = \sigma/\varepsilon_o$

Answer Given By Candidate:**C**

Question ID:101409
Topic Name: Physics-Section A
Question:

Match **List - I** with **List - II**.

	List - I		**List - II**
(A)	AC generator	(I)	Detects the presence of current in the circuit
(B)	Galvanometer	(II)	Converts mechanical energy into electrical energy
(C)	Transformer	(III)	Works on the principle of resonance in AC circuit
(D)	Metal detector	(IV)	Changes an alternating voltage for smaller or greater value

Choose the **correct** answer from the options given below :

A (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

B (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

C (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

D (A) - (III), (B) - (I), (C) - (II), (D) - (IV)

Answer Given By Candidate:**B**

Question ID:101410
Topic Name: Physics-Section A
Question:

A long straight wire with a circular cross-section having radius R, is carrying a steady current I. The current I is uniformly distributed across this cross-section. Then the variation of magnetic field due to current I with distance r (r < R) from its centre will be :

A $B \propto r^2$

B $B \propto r$

C $B \propto \frac{1}{r^2}$

D $B \propto \frac{1}{r}$

Answer Given By Candidate:**B**

Question ID:101411
Topic Name: Physics-Section A
Question:

If wattless current flows in the AC circuit, then the circuit is :

A Purely Resistive circuit

B Purely Inductive circuit

C LCR series circuit

D RC series circuit only

Answer Given By Candidate:**A**

Question ID:101412
Topic Name: Physics-Section A
Question:

The electric field in an electromagnetic wave is given by $E = 56.5 \sin \omega(t - x/c)\ NC^{-1}$. Find the intensity of the wave if it is propagating along x-axis in the free space.
(Given $\epsilon_0 = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$)

A $5.65\ Wm^{-2}$

B $4.24\ Wm^{-2}$

C $1.9 \times 10^{-7} Wm^{-2}$

D

$56.5\ Wm^{-2}$

Answer Given By Candidate:C

Question ID:101413
Topic Name: Physics-Section A
Question:

The two light beams having intensities I and 9I interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\pi/2$ at point P and π at point Q. Then the difference between the resultant intensities at P and Q will be :

A $2\ I$

B $6\ I$

C $5\ I$

D $7\ I$

Answer Given By Candidate:B

Question ID:101414
Topic Name: Physics-Section A
Question:

A light wave travelling linearly in a medium of dielectric constant 4, incidents on the horizontal interface separating medium with air. The angle of incidence for which the total intensity of incident wave will be reflected back into the same medium will be :
(Given : relative permeability of medium $\mu_r = 1$)

A $10°$

B $20°$

C $30°$

D $60°$

Answer Given By Candidate:Not Attempted

Question ID:101415
Topic Name: Physics-Section A
Question:

Given below are two statements :

Statement I : Davisson-Germer experiment establishes the wave nature of electrons.

Statement II : If electrons have wave nature, they can interfere and show diffraction.

In the light of the above statements choose the **correct answer** from the option given below :

A Both **Statement I** and **Statement II** are true.

B Both **Statement I** and **Statement II** are false.

C **Statement I** is true but **Statement II** is false.

D **Statement I** is false but **Statement II** is true.

Answer Given By Candidate:A

Question ID:101416
Topic Name: Physics-Section A
Question:

The ratio for the speed of the electron in the 3rd orbit of He^{+} to the speed of the electron in the 3rd orbit of hydrogen atom will be :

A 1 : 1

B 1 : 2

C 4 : 1

D 2 : 1

Answer Given By Candidate:**C**

Question ID:101417
Topic Name: Physics-Section A
Question:

The photodiode is used to detect the optical signals. These diodes are preferably operated in reverse biased mode because :

A fractional change in majority carriers produce higher forward bias current

B fractional change in majority carriers produce higher reverse bias current

C fractional change in minority carriers produce higher forward bias current

D fractional change in minority carriers produce higher reverse bias current

Answer Given By Candidate:**C**

Question ID:101418
Topic Name: Physics-Section A
Question:

A signal of 100 THz frequency can be transmitted with maximum efficiency by :

A Coaxial cable

B Optical fibre

C Twisted pair of copper wires

D Water

Answer Given By Candidate:**B**

Question ID:101419
Topic Name: Physics-Section A
Question:

The difference of speed of light in the two media A and B $(v_A - v_B)$ is 2.6×10^7 m/s. If the refractive index of medium B is 1.47, then the ratio of refractive index of medium B to medium A is : (Given : speed of light in vacuum $c = 3 \times 10^8$ ms^{-1})

A 1.303

B 1.318

C 1.13

D 0.12

Answer Given By Candidate:D

Question ID:101420
Topic Name: Physics-Section A
Question:

A teacher in his physics laboratory allotted an experiment to determine the resistance (G) of a galvanometer. Students took the observations for $\frac{1}{3}$ deflection in the galvanometer. Which of the below is **true** for measuring value of G ?

A $\frac{1}{3}$ deflection method cannot be used for determining the resistance of the galvanometer.

B $\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistance(s).

C $\frac{1}{3}$ deflection method can be used and in this case, the G equals to three times the value of shunt resistance(s).

D $\frac{1}{3}$ deflection method can be used and in this case the G value equals to the shunt resistance(s).

Answer Given By Candidate:Not Attempted

Question ID:101421
Topic Name: Physics-Section B
Question:

A uniform chain of 6 m length is placed on a table such that a part of its length is hanging over the edge of the table. The system is at rest. The co-efficient of static friction between the chain and the surface of the table is 0.5, the maximum length of the chain hanging from the table is _____m.

Answer Given By Candidate:Not Attempted

Question ID:101422
Topic Name: Physics-Section B
Question:

A 0.5 kg block moving at a speed of 12 ms^{-1} compresses a spring through a distance 30 cm when its speed is halved. The spring constant of the spring will be _____ Nm^{-1}.

Answer Given By Candidate:60

Question ID:101423
Topic Name: Physics-Section B

The velocity of upper layer of water in a river is 36 kmh^{-1}. Shearing stress between horizontal layers of water is 10^{-3} Nm^{-2}. Depth of the river is _____ m. (Co-efficient of viscosity of water is 10^{-2} Pa.s)

Question:

Answer Given By Candidate:**Not Attempted**

Question ID:101424
Topic Name: Physics-Section B

A steam engine intakes 50 g of steam at 100°C per minute and cools it down to 20°C. If latent heat of vaporization of steam is 540 cal g^{-1}, then the heat rejected by the steam engine per minute is _____ $\times 10^3$ cal.

(Given : specific heat capacity of water : 1 cal g^{-1} $°C^{-1}$)

Question:

Answer Given By Candidate:**Not Attempted**

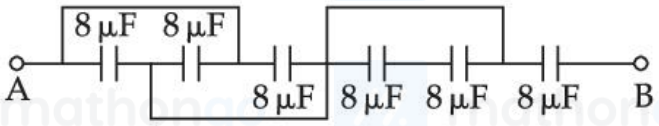
Question ID:101425
Topic Name: Physics-Section B
Question:

The first overtone frequency of an open organ pipe is equal to the fundamental frequency of a closed organ pipe. If the length of the closed organ pipe is 20 cm. The length of the open organ pipe is _____ cm.

Answer Given By Candidate:**Not Attempted**

Question ID:101426
Topic Name: Physics-Section B
Question:

The equivalent capacitance between points A and B in below shown figure will be _____ μF.



Answer Given By Candidate:**12**

Question ID:101427
Topic Name: Physics-Section B
Question:

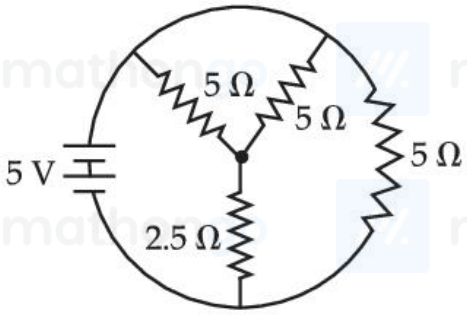
A resistor develops 300 J of thermal energy in 15 s, when a current of 2 A is passed through it. If the current increases to 3 A, the energy developed in 10 s is _____ J.

Answer Given By Candidate:**450**

Question ID:101428
Topic Name: Physics-Section B

Question:

The total current supplied to the circuit as shown in figure by the 5 V battery is _____ A.



Answer Given By Candidate:**4**

Question ID:101429
Topic Name: Physics-Section B
Question:

The current in a coil of self inductance 2.0 H is increasing according to $I = 2\sin(t^2)$ A. The amount of energy spent during the period when current changes from 0 to 2 A is _____ J.

Answer Given By Candidate:**Not Attempted**

Question ID:101430
Topic Name: Physics-Section B
Question:

A force on an object of mass 100 g is $\left(10\,\hat{i} + 5\,\hat{j}\right)$ N. The position of that object at t = 2 s is $\left(a\,\hat{i} + b\,\hat{j}\right)$ m after starting from rest. The value of $\frac{a}{b}$ will be _____.

Answer Given By Candidate:**2**

Question ID:101431
Topic Name: Chemistry-Section A
Question:

Bonding in which of the following diatomic molecule(s) become(s) stronger, on the basis of MO Theory, by removal of an electron ?

(A) NO
(B) N_2
(C) O_2
(D) C_2
(E) B_2

Choose the **most appropriate** answer from the options given below :

A (A), (B), (C) only

B (B), (C), (E) only

C (A), (C) only

D (D) only

Answer Given By Candidate:Not Attempted

Question ID:101432
Topic Name: Chemistry-Section A
Question: **Incorrect** statement for Tyndall effect is :

A The refractive indices of the dispersed phase and the dispersion medium differ greatly in magnitude.

B The diameter of the dispersed particles is much smaller than the wavelength of the light used.

C During projection of movies in the cinemas hall, Tyndall effect is noticed.

D It is used to distinguish a true solution from a colloidal solution.

Answer Given By Candidate:B

Question ID:101433
Topic Name: Chemistry-Section A
Question: The pair, in which ions are isoelectronic with Al^{3+} is :

A Br^- and Be^{2+}

B Cl^- and Li^+

C S^{2-} and K^+

D O^{2-} and Mg^{2+}

Answer Given By Candidate:D

Question ID:101434
Topic Name: Chemistry-Section A
Question:
Leaching of gold with dilute aqueous solution of NaCN in presence of oxygen gives complex [A], which on reaction with zinc forms the elemental gold and another complex [B]. [A] and [B], respectively are :

A $[Au(CN)_4]^-$ and $[Zn(CN)_2(OH)_2]^{2-}$

B $[Au(CN)_2]^-$ and $[Zn(OH)_4]^{2-}$

C $[Au(CN)_2]^-$ and $[Zn(CN)_4]^{2-}$

D $[Au(CN)_4]^{2-}$ and $[Zn(CN)_6]^{4-}$

Answer Given By Candidate:B

Question ID:101435
Topic Name: Chemistry-Section A

Number of electron deficient molecules among the following PH_3, B_2H_6, CCl_4, NH_3, LiH and BCl_3 is

Question:

A 0

B 1

C 2

D 3

Answer Given By Candidate:**C**

Question ID:101436
Topic Name: Chemistry-Section A
Question:

Which one of the following alkaline earth metal ions has the highest ionic mobility in its aqueous solution ?

A Be^{2+}

B Mg^{2+}

C Ca^{2+}

D Sr^{2+}

Answer Given By Candidate:**A**

Question ID:101437
Topic Name: Chemistry-Section A
Question:

White precipitate of AgCl dissolves in aqueous ammonia solution due to formation of :

A $[Ag(NH_3)_4]Cl_2$

B $[Ag(Cl)_2(NH_3)_2]$

C $[Ag(NH_3)_2]Cl$

D $[Ag(NH_3)Cl]Cl$

Answer Given By Candidate:**A**

Question ID:101438
Topic Name: Chemistry-Section A
Question:

Cerium (IV) has a noble gas configuration. Which of the following is **correct** statement about it ?

A It will not prefer to undergo redox reactions.

B It will prefer to gain electron and act as an oxidizing agent

C It will prefer to give away an electron and behave as reducing agent

D

It acts as both, oxidizing and reducing agent.

Answer Given By Candidate:**B**

Question ID:101439
Topic Name: Chemistry-Section A
Question: Among the following, which is the strongest oxidizing agent ?

A Mn^{3+}

B Fe^{3+}

C Ti^{3+}

D Cr^{3+}

Answer Given By Candidate:**C**

Question ID:101440
Topic Name: Chemistry-Section A
Question: The eutrophication of water body results in :

A loss of Biodiversity.

B breakdown of organic matter.

C increase in biodiversity.

D decrease in BOD.

Answer Given By Candidate:**A**

Question ID:101441
Topic Name: Chemistry-Section A
Question:
Phenol on reaction with dilute nitric acid, gives two products. Which method will be most efficient for large scale separation ?

A Chromatographic separation

B Fractional Crystallisation

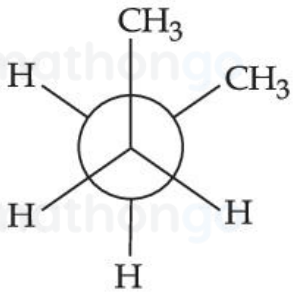
C Steam distillation

D Sublimation

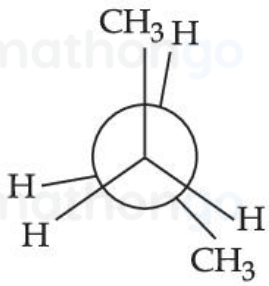
Answer Given By Candidate:**C**

Question ID:101442
Topic Name: Chemistry-Section A
Question:
In the following structures, which one is having staggered conformation with maximum dihedral angle ?

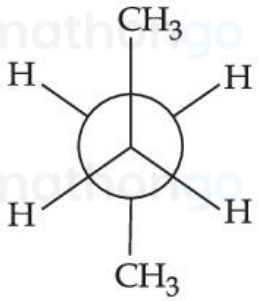
A



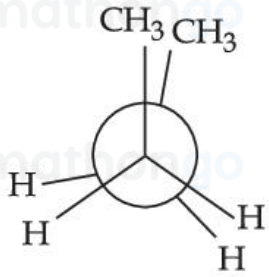
B



C



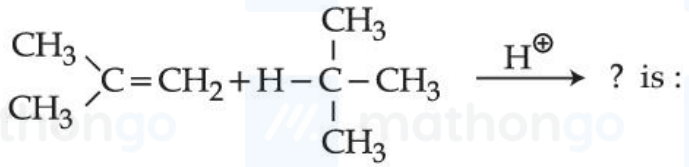
D



Answer Given By Candidate:**Not Attempted**

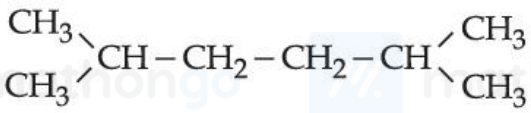
Question ID:101443
Topic Name: Chemistry-Section A

The product formed in the following reaction.

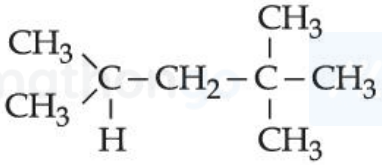


Question:

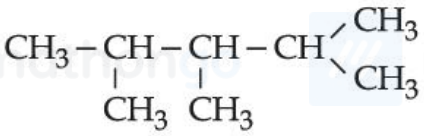
A



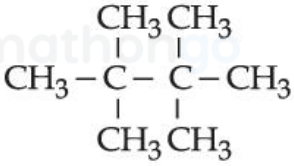
B



C



D



Answer Given By Candidate:**C**

Question ID:101444
Topic Name: Chemistry-Section A

Question: The IUPAC name of ethylidene chloride is :

A 1-Chloroethene

B 1-Chloroethyne

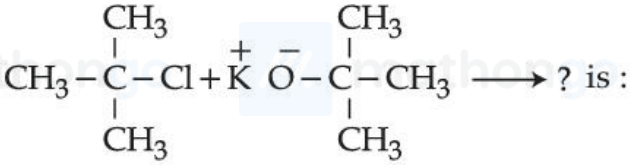
C 1,2-Dichloroethane

D 1,1-Dichloroethane

Answer Given By Candidate:**A**

Question ID:101445
Topic Name: Chemistry-Section A

Question: The major product in the reaction



A *t*-Butyl ethyl ether

B 2,2-Dimethyl butane

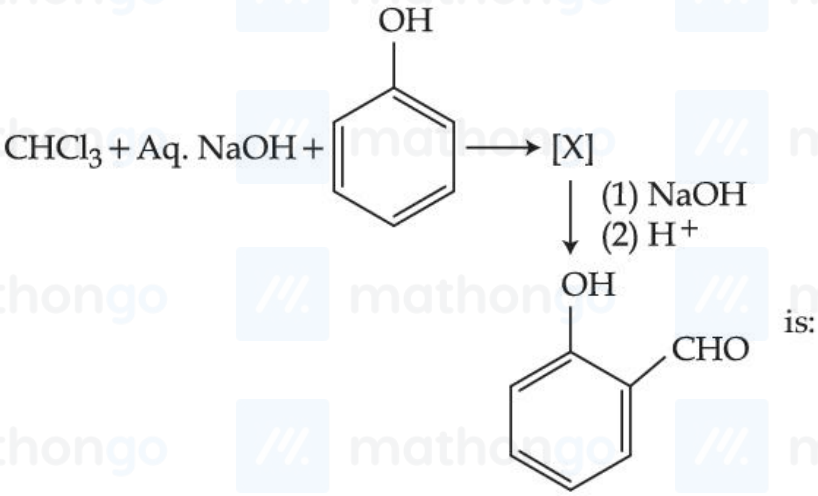
C 2-Methyl pent-1-ene

D 2-Methyl prop-1-ene

Answer Given By Candidate:**Not Attempted**

Question ID:101446
Topic Name: Chemistry-Section A

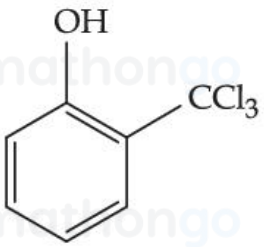
The intermediate X, in the reaction :



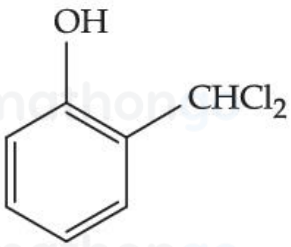
is:

Question:

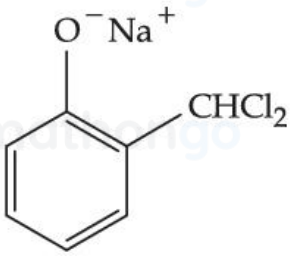
A



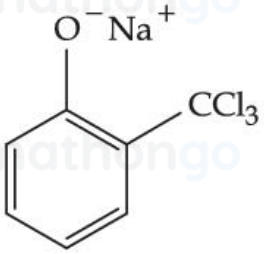
B



C



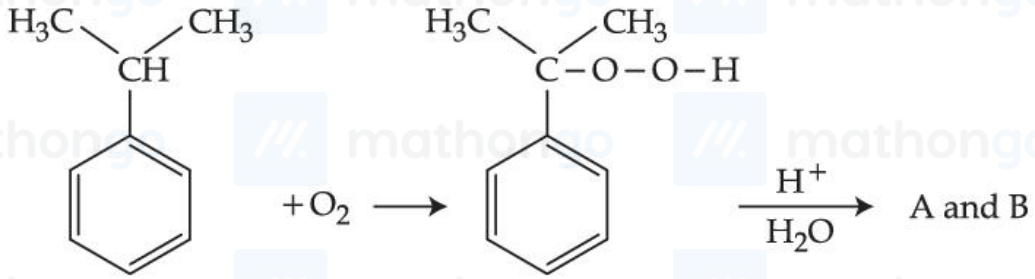
D



Answer Given By Candidate:**C**

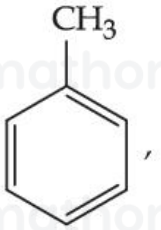
Question ID:101447
Topic Name: Chemistry-Section A

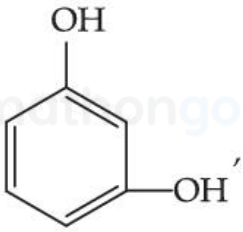
In the following reaction :

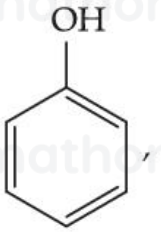


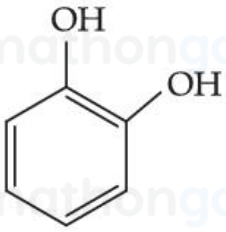
The compounds A and B respectively are :

Question:

A  , CH_3COOH

B  , CH_3COCH_3

C  , CH_3COCH_3

D  , CH_3COCH_3

Answer Given By Candidate:**A**

Question ID:101448
Topic Name: Chemistry-Section A
Question:

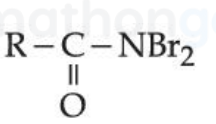
The reaction of $R-\underset{\underset{O}{\|}}{C}-NH_2$ with bromine and KOH gives RNH_2 as the end product. Which one of the following is the intermediate product formed in this reaction ?

A $R-\underset{\underset{O}{\|}}{C}-NH-Br$

B $R-NH-Br$

C

D $R-N=C=O$

D 

Answer Given By Candidate:**C**

Question ID:101449
Topic Name: Chemistry-Section A
Question:

Using very little soap while washing clothes, does not serve the purpose of cleaning of clothes, because :

A soap particles remain floating in water as ions.

B the hydrophobic part of soap is not able to take away grease.

C the micelles are not formed due to concentration of soap, below its CMC value.

D colloidal structure of soap in water is completely disturbed.

Answer Given By Candidate:**C**

Question ID:101450
Topic Name: Chemistry-Section A
Question: Which one of the following is an example of artificial sweetner ?

A Bithional

B Alitame

C Salvarsan

D Lactose

Answer Given By Candidate:**B**

Question ID:101451
Topic Name: Chemistry-Section B
Question:

The number of N atoms in 681 g of $C_7H_5N_3O_6$ is $x\times10^{21}$. The value of x is _____.
$(N_A=6.02\times10^{23}\ mol^{-1})$ (Nearest Integer)

Answer Given By Candidate:**37**

Question ID:101452
Topic Name: Chemistry-Section B
Question:

The distance between Na^+ and Cl^- ions in solid NaCl of density 43.1 g cm^{-3} is _____ $\times10^{-10}$ m. (Nearest Integer)
(Given : $N_A=6.02\times10^{23}\ mol^{-1}$)

Answer Given By Candidate:**Not Attempted**

Question ID:101453
Topic Name: Chemistry-Section B

Question:

The longest wavelength of light that can be used for the ionisation of lithium atom (Li) in its ground state is $x \times 10^{-8}$ m. The value of x is _____. (Nearest Integer)

(Given : Energy of the electron in the first shell of the hydrogen atom is -2.2×10^{-18} J ; $h = 6.63 \times 10^{-34}$ Js and $c = 3 \times 10^{8}$ ms^{-1})

Answer Given By Candidate:**308**

Question ID:101454
Topic Name: Chemistry-Section B
Question:

The standard entropy change for the reaction

$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$ is -550 J K^{-1} at 298 K.

[Given : The standard enthalpy change for the reaction is -165 kJ mol^{-1}]. The temperature in K at which the reaction attains equilibrium is _____. (Nearest Integer)

Answer Given By Candidate:**Not Attempted**

Question ID:101455
Topic Name: Chemistry-Section B
Question:

1 L aqueous solution of H_2SO_4 contains 0.02 m mol H_2SO_4. 50% of this solution is diluted with deionized water to give 1 L solution (A). In solution (A), 0.01 m mol of H_2SO_4 are added. Total m mols of H_2SO_4 in the final solution is _____ $\times 10^3$ m mols.

Answer Given By Candidate:**Not Attempted**

Question ID:101456
Topic Name: Chemistry-Section B
Question:

The standard free energy change (ΔG°) for 50% dissociation of N_2O_4 into NO_2 at 27°C and 1 atm pressure is $-x$ J mol^{-1}. The value of x is _____. (Nearest Integer)

[Given : $R = 8.31$ J K^{-1} mol^{-1}, $\log 1.33 = 0.1239$ $\ln 10 = 2.3$]

Answer Given By Candidate:**Not Attempted**

Question ID:101457
Topic Name: Chemistry-Section B
Question:

In a cell, the following reactions take place

$$Fe^{2+} \rightarrow Fe^{3+} + e^- \qquad E^\circ_{Fe^{3+}/Fe^{2+}} = 0.77 \text{ V}$$

$$2I^- \rightarrow I_2 + 2e^- \qquad E^\circ_{I_2/I^-} = 0.54 \text{ V}$$

The standard electrode potential for the spontaneous reaction in the cell is $x \times 10^{-2}$ V 298 K. The value of x is _____. (Nearest Integer)

Answer Given By Candidate:**23**

Question ID:101458
Topic Name: Chemistry-Section B

Question:

For a given chemical reaction

$\gamma_1 A + \gamma_2 B \rightarrow \gamma_3 C + \gamma_4 D$

Concentration of C changes from 10 mmol dm^{-3} to 20 mmol dm^{-3} in 10 seconds. Rate of appearance of D is 1.5 times the rate of disappearance of B which is twice the rate of disappearance A. The rate of appearance of D has been experimentally determined to be 9 mmol dm^{-3} s^{-1}. Therefore the rate of reaction is _____ mmol dm^{-3} s^{-1}. (Nearest Integer)

Answer Given By Candidate:**Not Attempted**

Question ID:101459
Topic Name: Chemistry-Section B
Question:

If $[Cu(H_2O)_4]^{2+}$ absorbs a light of wavelength 600 nm for d-d transition, then the value of octahedral crystal field splitting energy for $[Cu(H_2O)_6]^{2+}$ will be _____ $\times 10^{-21}$ J. [Nearest integer]

(Given : $h = 6.63 \times 10^{-34}$ Js and $c = 3.08 \times 10^{8}$ ms^{-1})

Answer Given By Candidate:**3**

Question ID:101460
Topic Name: Chemistry-Section B
Question:

Number of grams of bromine that will completely react with 5.0 g of pent-1-ene is _____ $\times 10^{-2}$ g. (Atomic mass of Br = 80 g/mol) [Nearest integer]

Answer Given By Candidate:**Not Attempted**